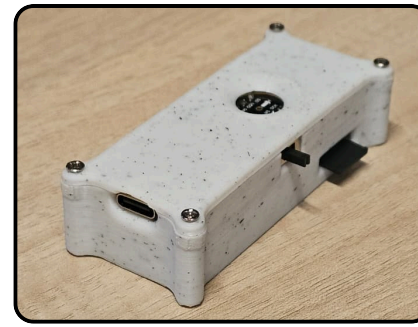




DEEPLLOGIX

An End-to-End Voice Capture and Processing System Using IoT and AI

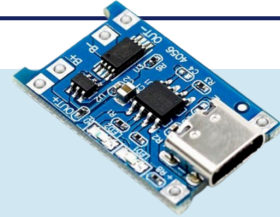
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CPG-142

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01. Introduction



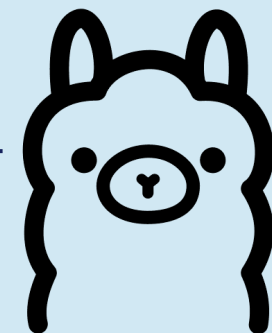
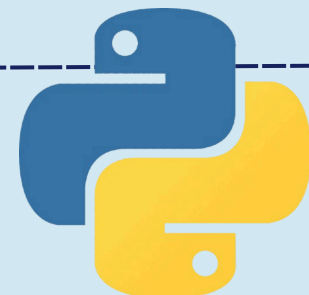
- Spoken interactions often remain undocumented, resulting in information loss and poor recall.
- DeepLogix captures short voice recordings using a compact hardware device and converts them into structured digital data via backend speech processing.
- The processed information is presented through a mobile application for review and summarization, emphasizing simplicity, system integration, and privacy.

02. Problem Statement

- Important conversations are often not captured in a structured or retrievable form.
- Existing voice recording tools store only raw audio, while manual note-taking is time-consuming and incomplete.
- As a result, users lack a simple and reliable way to review conversations, extract insights, and reflect on communication patterns.

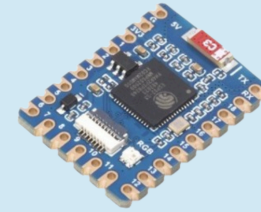


03. Objective

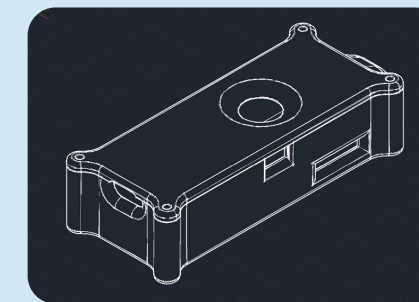
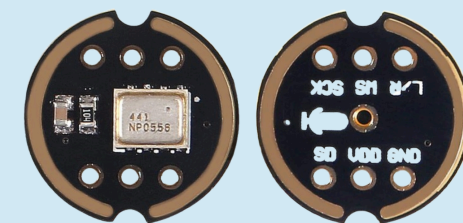


- To design a compact hardware device for capturing voice conversations and storing audio in standard WAV format.
- To transmit recorded audio over Wi-Fi to a backend server for speech-to-text conversion and structured data processing.
- To present transcriptions, summaries, and insights through a mobile application, demonstrating end-to-end integration of hardware, backend, database, and AI components.

04. Hardware Overview

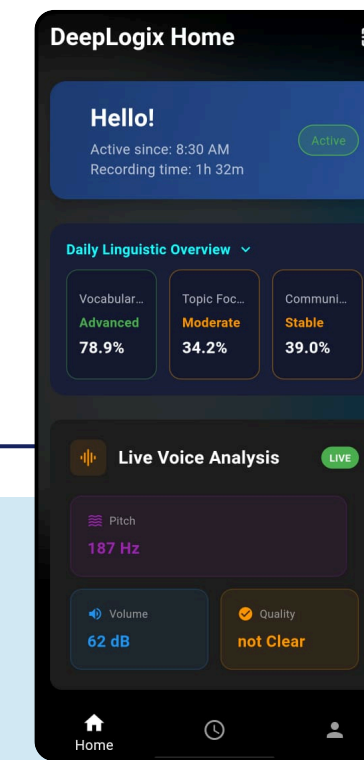


- ESP32-S3 Mini manages audio capture, local storage, and Wi-Fi communication, while the INMP441 I2S microphone provides clear digital voice input.
- Recorded audio is stored locally in WAV format on a micro SD card, with a single-cell Li-ion battery and Type-C charging enabling portable operation.
- The hardware acts as a lightweight voice acquisition unit, handling only recording and transmission, while all processing is performed on the software side.



06. Results and Outcome

- Clear and consistent voice capture was achieved using the ESP32-S3 and INMP441 I2S microphone, with audio stored reliably in WAV format.
- Recorded audio was successfully transmitted over Wi-Fi and converted into structured text using backend speech recognition.
- Transcriptions were synchronized with the mobile application in near real time and presented in a time-stamped, organized format for easy review.



05. System Architecture and WorkFlow

- Voice is captured using an INMP441 I2S microphone via the ESP32-S3, stored as WAV files, and transmitted to a backend server over Wi-Fi using HTTP.
- A Flask-based backend performs speech-to-text conversion with noise handling, stores transcriptions in Firebase Firestore, and enables real-time synchronization.
- Local AI models generate summaries and insights, which are presented through a Flutter-based mobile application for structured review and user interaction.

